

AARIZ ZAFAR, Bangalore, India

Machine Learning Engineer specializing in Computer Vision, Predictive Analytics & Cloud AI

Email: aariz.zafar01@gmail.com | Phone: +91 9353707832 | GitHub: github.com/AarizZafar | LinkedIn: linkedin.com/in/aariz-zafar

CAREER EXECUTIVE SUMMARY

Results-driven Computer Science Engineering graduate with 1+ year of hands-on industry experience in Machine Learning, Data Science, and Cloud Computing. Specialized in developing AI/ML solutions for predictive maintenance, synthetic data generation using physics-based simulation, and computer vision applications for industrial safety. Proficient in Python, PyTorch, PySpark, and Azure cloud technologies, with three Microsoft Azure certifications. Strong foundation in building and deploying scalable ML pipelines that enhance operational efficiency and solve real-world business challenges.

QUALIFICATIONS

Bachelor of Technology (B. Tech)
Computer Science and Engineering
CGPA: 8.66/10.0 (2025)
Vellore Institute of Technology AP, India

CERTIFICATION

- Microsoft Azure AZ-500 Security Engineer Associate
- Microsoft Azure AZ-104 Administrator Associate
- Microsoft Azure AZ-900 Fundamentals
- Machine Learning– Boston Training Academy | AI & ML Program– Google Developer (Smart Internz)
- Bigdata engineering with GCP and Azure - Udemy
- BASH Shell scripting certification - Udemy

SKILL

1. **Programming Languages:**
 - Python, C++, SQL, Java, R, Golang (Application development, scripting, data processing, algorithm implementation)
2. **Machine Learning & AI:**
 - PyTorch, TensorFlow, Keras, Scikit-learn.
 - NLP and LLM using Hugging Face Transformers.
 - Retrieval-Augmented Generation (RAG) using LangChain and vector databases
 - Agentic AI using LangChain and LangGraph
 - Computer Vision for object detection specially PPE violation in factories.
3. **Data Analysis & Visualization:**
 - NumPy, Pandas, OpenCV (Data manipulation, feature engineering, image processing, exploratory analysis)
4. **Big Data & Data Engineering:**
 - Apache Spark (PySpark), Apache Kafka, Apache Airflow, Hadoop (HDFS, MapReduce, YARN) (Distributed data processing, streaming, workflow orchestration, ETL pipelines)
5. **Cloud & MLOps:**
 - Microsoft Azure (Data Factory, Databricks, Synapse Analytics, ADLS Gen2), Google Cloud Platform (Dataproc), Docker (Cloud-native data platforms, ML pipeline deployment, containerization)
6. **Databases:**
 - PostgreSQL, MySQL, MongoDB (Relational & NoSQL database design, querying, data management)
7. **Domain Expertise:**
 - Predictive Maintenance, Fault Detection & Diagnosis (FDD), Building Energy Simulation.
 - Industrial Computer Vision for PPE Compliance Monitoring
 - Sustainability & Energy Optimization using Machine Learning.

WORK EXPERIENCE AND PROJECTS DETAILS

1. **Trinity Mobility - AI Engineer , Bangalore, India | Mar 2026 – Present**
 - **Predictive Maintenance:** Designed and developed a machine learning-based Fault Detection and Diagnosis (FDD) framework for HVAC equipment (chillers, AHUs, FCUs) using Python, Scikit-learn, and time-series anomaly detection models, enabling early fault identification from synthetic sensor data.
 - **Synthetic Data Generation:** Built physics-based simulation models using EnergyPlus and Modelica to generate 40+ refrigerant degradation fault scenarios, creating high-quality synthetic datasets to overcome limited real-world training data.
 - **ML Pipeline Development:** Developed end-to-end data pipelines for HVAC sensor data ingestion, preprocessing, feature engineering, and model training, integrating simulation outputs with supervised learning models for equipment health monitoring.

2. LivNSense – Software Engineer (Intern + Full time employee), Bangalore, India | Dec 2024 – Mar 2026

- **Energy Optimization:** Developed an end-to-end machine learning pipeline for asphalt plant optimization using exploratory data analysis, statistical modelling, and anomaly detection on large-scale manufacturing time-series data, improving operational efficiency by **10%**.
- **Retrieval-Augmented Generation (RAG):** Designed and deployed a domain-specific RAG solution using LangChain, Hugging Face Transformers, and vector embeddings, reducing engineering knowledge retrieval effort by **30%**.
- **Computer Vision:** Enhanced an industrial PPE detection system by developing a high-performance C++ image preprocessing pipeline, improving object detection accuracy from **75% to 95%** for real-time monitoring of 100+ workers.

3. NETTECH - Software Engineering Intern, New Delhi, India, Oct 2023 – Dec 2023 (Academic Projects)

- **NLP & RAG System:** Developed a Retrieval-Augmented Generation (RAG) solution using LangChain and SBERT embeddings to enable natural language querying of network device documentation, reducing manual configuration lookup effort by **20%**.

INDUSTRY COLLABORATION

Represented LivNSense at Automation India Expo 2026, demonstrating AI-driven industrial automation solutions to industry partners

Reference: [Automation India Expo 2026](#) (LinkedIn)

AWARDS & RECOGNITION

Recipient of the Best Employee Award for outstanding contribution to AI-based industrial optimization projects.